FIT5196 A1 — Group030

Focused exploratory analysis of reconciled 2018 commerce and operations data

Prepared from six standardised relational tables

Assessed report: 7 figures, 10 findings, 5 future ML questions

1. Context and preparation assurance

The workflow integrates JSON commerce and XML operations exports into six analysis-ready tables at their published grains. Material decisions: structured parsing precedes regex; identifiers retain case and leading zeros; duplicate records are compared after normalisation and reconciled by business key; monetary arithmetic follows rounded line revenue, included GST, discount then delivery; multilingual cleaned reviews are preserved separately from Latin-only analysis (MAP-*).

Validation evidence: all six schemas and field orders pass (VAL-SCHEMA-*); primary and foreign keys are complete and unique (VAL-PK-* / VAL-FK-*); no normalised cross-source conflicts remain (VAL-FLOW-01); arithmetic differences are within $0.01 (VAL-ARITH-*); temporal checks pass at the published date/timestamp precision (VAL-TIME-*); and literal NaN plus multilingual indicators pass (VAL-TEXT-*). Join assertions in the EDA preserve the left-table grain.

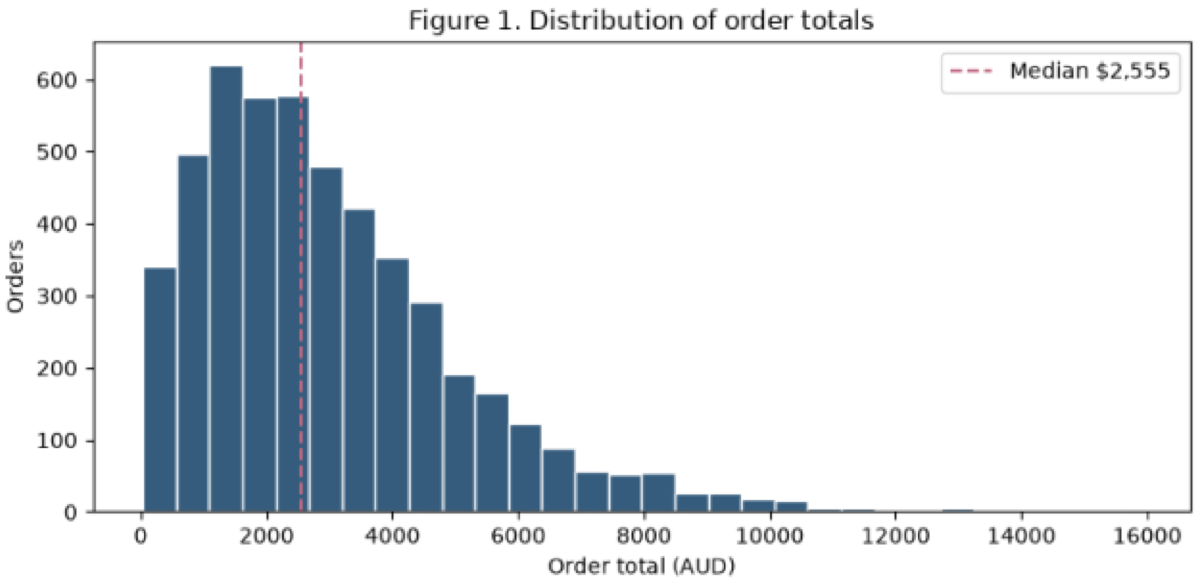


Figure 1. Distribution of order totals
Median $2,555
Orders
600
500
400
300
200
100
0
0
2000
4000
6000
8000
10000
12000
14000
16000
Order total (AUD)

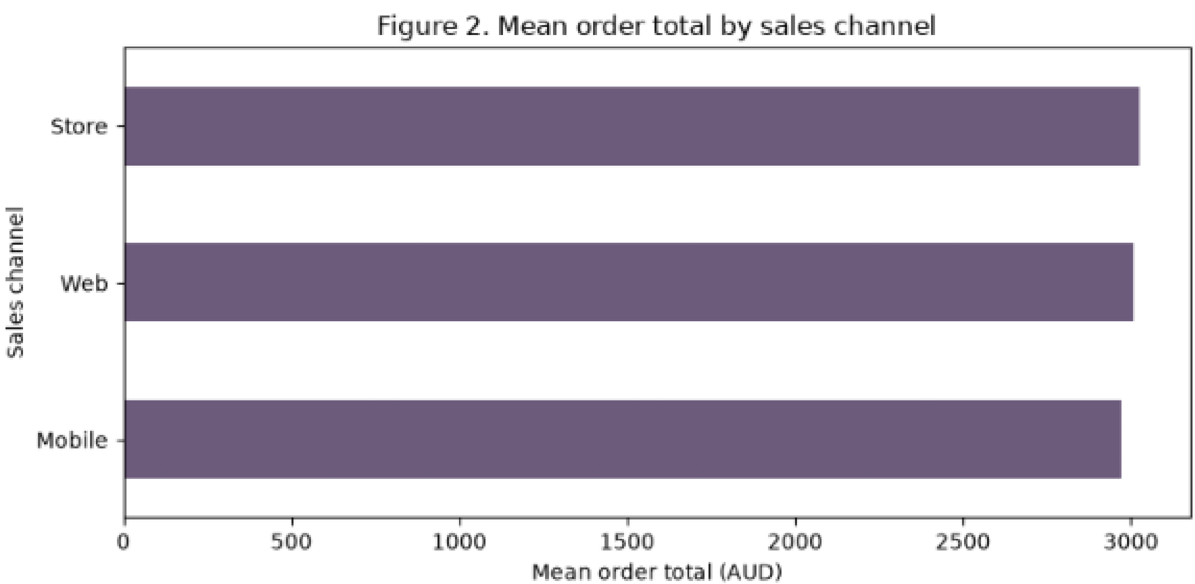


Figure 2. Mean order total by sales channel
Sales channel
Store
Web
Mobile
0
500
1000
1500
2000
2500
3000
Mean order total (AUD)

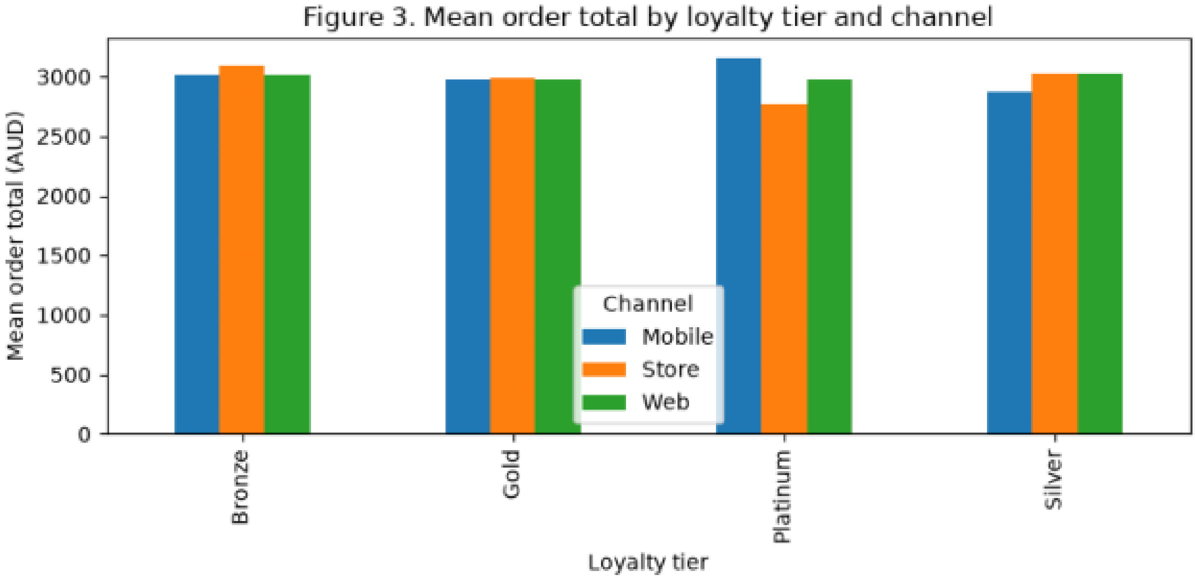


Figure 3. Mean order total by loyalty tier and channel
Mean order total (AUD)
3000
2500
2000
1500
1000
500
0
Channel
Mobile
Store
Web
Bronze
Gold
Platinum
Silver
Loyalty tier

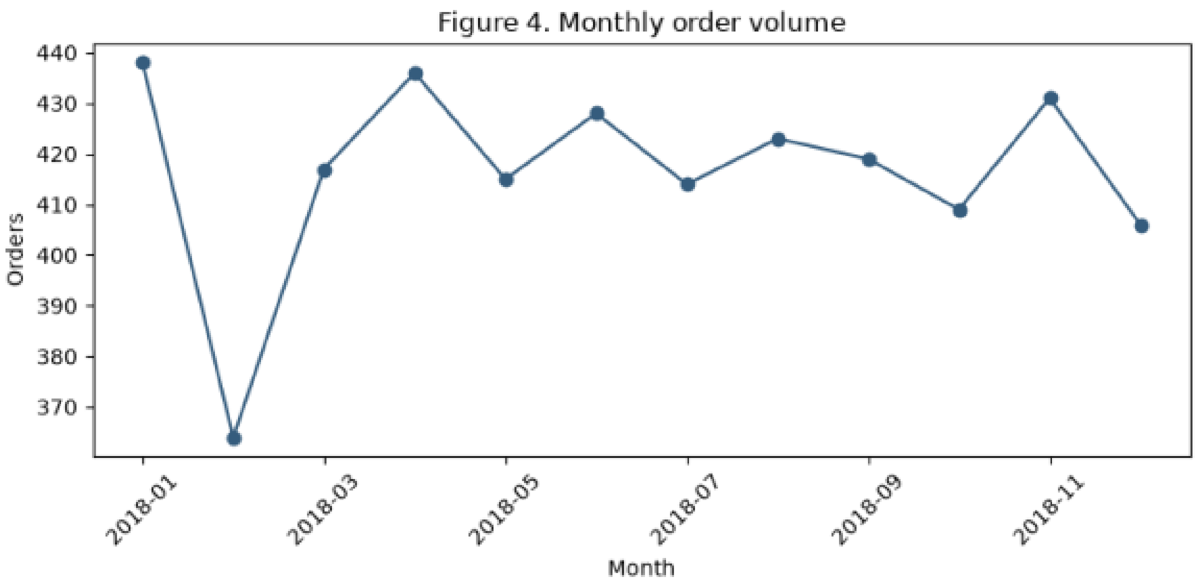


Figure 4. Monthly order volume
Orders
440
430
420
410
400
390
380
370
2018-01
2018-03
2018-05
2018-07
2018-09
2018-11
Month

Figure 5. Review behaviour

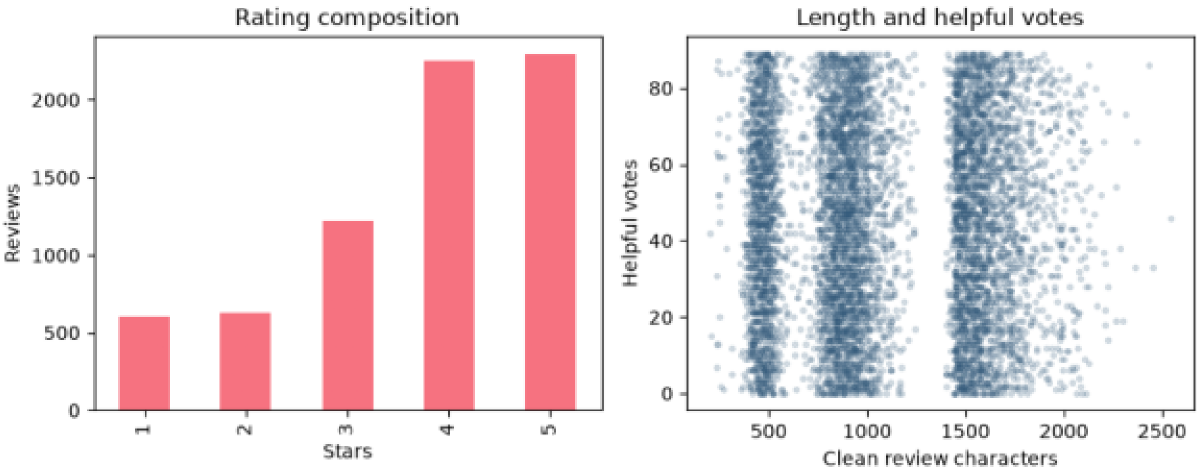
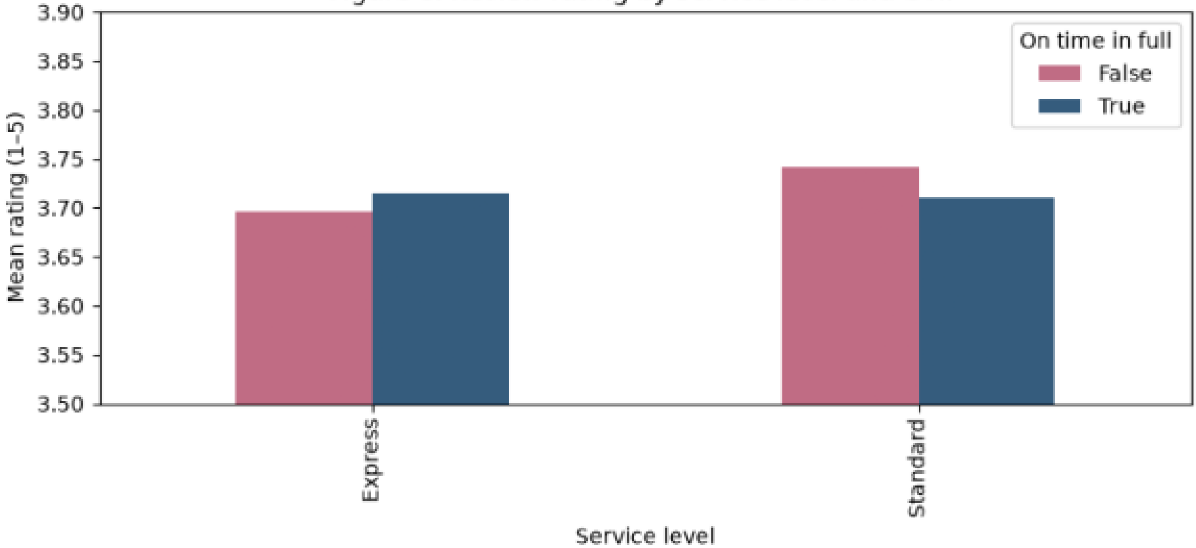


Figure 6. Review rating by service level and OTIF



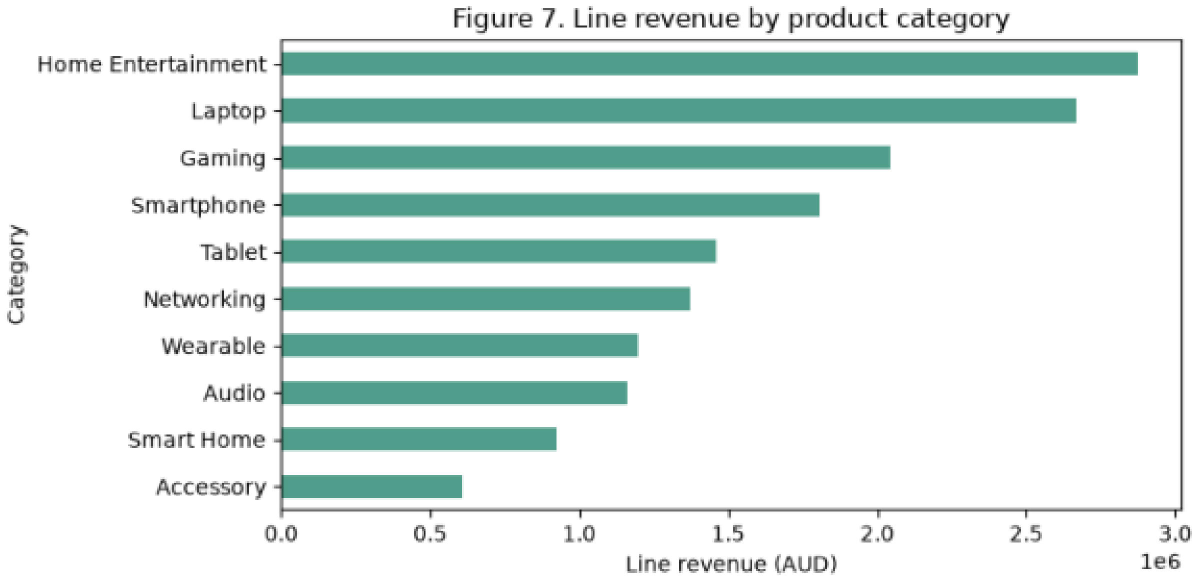


Figure 7. Line revenue by product category
Home Entertainment
Laptop
Gaming
Smartphone
Tablet
Networking
Wearable
Audio
Smart Home
Accessory
Category
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1e6
Line revenue (AUD)

3. Ten evidence-based findings

1. Figure 1: Across 5,000 orders, the median order total was $2,555.44; the broad distribution makes median-based planning safer than relying only on the mean. Product mix may explain the spread, so investigate category-adjusted values.

2. Figure 1: Total 2018 order revenue was $15,001,855.75 across 5,000 orders. This establishes commercial scale but excludes margin and returns; pair revenue with product cost before profitability decisions.

3. Figure 2: Mean order values were similar across Web, Store and Mobile relative to within-channel dispersion. Channel is therefore a weak standalone basis for targeting; uncertainty should be quantified before reallocating spend.

4. Figure 3: Loyalty-tier/channel cells show only modest mean-value separation at the order grain. Unequal cell sizes and repeated customers can confound the pattern; use customer-level sensitivity analysis before tier policy changes.

5. Figure 4: Monthly order volume ranged from 364 to 438 orders. The variation may reflect calendar length or promotions rather than seasonality; multiple years and daily rates are needed for staffing forecasts.

6. Figure 5: Mean rating was 3.71/5 across 7,000 reviews, with four- and five-star reviews most common. Verified purchase is constant, so it cannot explain rating variation in this period.

7. Figure 5: Review length and helpful votes had correlation r=0.005 across 7,000 reviews—effectively no linear association. Nonlinear effects and review exposure remain plausible, so length should not be optimised alone.

8. Figure 5: 303 of 7,000 reviews contained non-Latin letters. Erasing them would systematically discard customer evidence; preserve multilingual text and audit language-specific performance.

9. Figure 6: 536 of 5,000 deliveries were not OTIF, yet mean ratings were close across OTIF groups. Product experience may dominate, and observational averages do not imply delivery has no causal effect.

10. Figure 7: Home Entertainment generated the highest line revenue ($2,876,082.60). Revenue is not profit and category prices differ; combine unit costs and quantities before assortment decisions.

4. Five future machine-learning questions

MLQ-1 — Classification: At order placement, predict failure to deliver OTIF to choose proactive intervention. Unit/target: order/OTIF failure. Predictors: warehouse, service level, distance, basket size, channel and order time available then. Use forward-chaining time split; evaluate PR-AUC and recall at intervention capacity. Exclude delivered date, delay reason and post-order tracking; audit suburb/service disparities.

MLQ-2 — Regression: Predict order total at session checkout for capacity and offer planning. Unit/target: order/final total. Use customer history, channel, season and pre-checkout basket features; temporal holdout and MAE. Coupon choice or final line totals can leak the target depending on decision time; monitor error by customer segment.

MLQ-3 — Classification: Predict whether a review will be 1–2 stars to prioritise service recovery. Unit/target: completed order/review rating class. Use order, product, delivery-service and customer history available before review. Temporal split; PR-AUC and calibrated precision. Never use review text, helpful votes or post-review fields; non-reviewers create selection bias.

MLQ-4 — Forecasting: Forecast weekly order counts by channel for staffing. Unit/target: channel-week/order count. Use lagged counts, calendar, season and known planned promotions; rolling-origin validation and WAPE. One year limits seasonal learning, and unknown future promotions/weather create deployment drift.

MLQ-5 — Clustering: Identify product demand/value profiles for assortment review. Unit/objective: product, clustered on pre-period or rolling line volume, revenue, price, cost and review summaries. Validate stability across time windows and silhouette plus business interpretability. Scaling choices dominate clusters; popularity exposure and sparse products can bias conclusions.

5. Limitations and conclusion

Limitations. The data cover one historical year and one retailer, so seasonality and external validity are limited. Associations are observational, repeated customers/products are not independent, and review analyses condition on customers who submitted reviews. Revenue is not profit; delivery-date fields are day-level; small language groups make subgroup estimates uncertain. No causal or model-performance claim is made.

Conclusion. The reconciled relational data show broad order-value variation, stable channel-level averages, commercially concentrated product-category revenue, high but imperfect OTIF performance and little raw relationship between review helpfulness and length. Operational prediction and forecasting are feasible next steps only with temporal evaluation, decision-time feature controls and subgroup monitoring.